Power-Bounded Weighted Composition Operators on Fock Spaces: Complete Compact and Zero-Free Classifications

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Status: substantial partial answer, likely valid, pending human review. For $1 \le p < \infty$ the problem is completely solved for arbitrary compact weights and for every zero-free weight. The only untreated class is simultaneously noncompact and zero-bearing.

1 Exact source problem

For $1 \le p < \infty$, write

$$\|f\|_p^p = \frac{p}{2\pi} \int_{\mathbb{C}} |f(z)|^p e^{-p|z|^2/2} \, dA(z)$$

for the standard Fock-space norm. Given entire u and $\varphi(z) = az + b$, the weighted composition operator is

$$W_{u,\varphi} f(z) = u(z) f(\varphi(z)).$$

Chalendar and Partington state, after their complete result for $|a| = 1$, that the case $|a| < 1$ is apparently unsolved [1]. Thus the target is:

Source problem. *Characterize the bounded operators $W_{u,\varphi}$ on $\mathcal{F}^p$ which are power bounded when $\varphi(z) = az + b$ and $|a| < 1$.*

See also [18] for a recent discussion of these results. Note that from [34] we have the useful inequality

$$\sqrt{M(w,\phi)} \le \|W_{w,\phi}\| \le \sqrt{M(w,\phi)}/|a|.$$

Thus for power-boundedness of $W_{w,\phi}$, it is possible to give a complete result for $|a| = 1$.

Theorem 6.7. *Suppose that $\phi(z) = az + b$ with $|a| = 1$, then $W_{w,\phi}$ is power-bounded on $\mathcal{F}^\nu$ if and only if $|w(0)| \le e^{-|b|^2/2}$.*

The case $|a| < 1$ is apparently unsolved, although this result for non-vanishing w (as in the case of semigroups) can be found in [18].

Theorem 6.8. *For $\phi(z) = az + b$ with $|a| < 1$ and w nonvanishing the operator $W_{w,\phi}$ is bounded on $\mathcal{F}^\nu$ if and only if*

The closest papers give the necessary condition $|u(b/(1-a))| \le 1$, a stronger sufficient condition, and an equivalence for compact zero-free weights [2, 3]. They leave a genuine gap even within the zero-free boundary class. The theorem below closes that gap and removes the zero-free assumption throughout the compact class.

2 Result

Let

$$\alpha = \frac{b}{1-a}$$

be the fixed point of φ.

Theorem 1 (compact and zero-free classification)**.** *Let $1 \le p < \infty$, $|a| < 1$, and suppose $W_{u,\varphi}$ is bounded on $\mathcal{F}^p$.*

1. *If $W_{u,\varphi}$ is compact, with no restriction on the zeros of u, then*

$$W_{u,\varphi} \text{ is power bounded} \iff |u(\alpha)| \le 1.$$

2. *Suppose u is zero-free. If $W_{u,\varphi}$ is compact or $a \notin \mathbb{R}$, the same equivalence holds.*
3. *Suppose u is zero-free, $W_{u,\varphi}$ is noncompact, and $a \in \mathbb{R} \setminus \{0\}$. Then the sharp criterion is*

$$W_{u,\varphi} \text{ is power bounded} \iff |u(\alpha)| \le |a|^{1/p}.$$

The real noncompact threshold is the new phenomenon. In the canonical Hilbert-space case its nth power contains a Gaussian dilation of norm comparable to $|a|^{-n/2}$, so the threshold is $|a|^{1/2}$ rather than the previously sufficient value $|a|$.

3 Centering and the compact case

For $c \in \mathbb{C}$, the Weyl translation

$$(U_c f)(z) = e^{\bar{c}z - |c|^2/2} f(z - c)$$

is an isometry of every $\mathcal{F}^p$. Direct calculation gives

$$U_\alpha^{-1} W_{u,\varphi} U_\alpha = W_{v,az}, \qquad v(z) = u(z+\alpha) e^{\overline{\alpha}(a-1)z}, \tag{1}$$

and, crucially, $v(0) = u(\alpha)$. We may therefore work at the fixed point 0.

Proposition 1. *If $W_{v,az}$ is compact on $\mathcal{F}^p$, then it is power bounded if and only if $|v(0)| \le 1$.*

Proof. For every n,

$$(W_{v,az}^n 1)(0) = v(0)^n,$$

so power boundedness forces $|v(0)| \le 1$.

We determine the nonzero spectral values directly. If $W_{v,az} f = \lambda f$, let m be the order of the first nonzero Taylor coefficient of f at 0. If $v(0) \neq 0$, comparison of the coefficient of z^m gives

$$\lambda = v(0) a^m. \tag{2}$$

If $v(0) = 0$, comparison of orders shows that no nonzero eigenvalue exists. Since the operator is compact, every nonzero spectral value is an eigenvalue. Consequently its spectral radius is $|v(0)|$.

If $|v(0)| < 1$, the powers tend to zero in norm. If $|v(0)| = 1$, the only peripheral spectral value is $v(0)$, corresponding to $m = 0$. It is semisimple: an eigenvector f for $v(0)$ satisfies $f(0) \neq 0$, while an equation $(W_{v,az} - v(0)I)g = f$ would give $0 = f(0)$ after evaluation at zero. The compact spectral decomposition is therefore $W_{v,az} = v(0)P + R$, where P is the spectral projection and $r(R) < 1$. Its powers are bounded. □

This proves part 1 of Theorem 1, including weights with arbitrary zero sets.

4 Zero-free weights and the Gaussian boundary

Assume now that v is zero-free. The standard boundedness criterion and Hadamard factorization give

$$v(z) = e^{P + Qz + Rz^2}, \qquad \beta := 1 - |a|^2, \qquad |R| \le \frac{\beta}{2}. \tag{3}$$

The operator is compact exactly when $|R| < \beta/2$. On the boundary $|R| = \beta/2$, boundedness forces either $Q = 0$ or

$$R = -\frac{\beta}{2} \frac{Q^2}{|Q|^2}. \tag{4}$$

These are the source's quoted structural conditions [2].

Put $c = v(0) = e^P$. The normalized powers have the explicit form

$$c^{-n} W_{v,az}^n f(z) = \exp\left(Q \frac{1 - a^n}{1 - a} z + R \frac{1 - a^{2n}}{1 - a^2} z^2 \right) f(a^n z). \tag{5}$$

The nonreal case

Suppose $a \notin \mathbb{R}$. At the noncompact boundary,

$$\left|\frac{R}{1-a^2}\right| = \frac{1-|a|^2}{2|1-a^2|} < \frac{1}{2}. \tag{6}$$

Thus the quadratic exponent in the normalized weight in (5) converges to a strictly subcritical Fock Gaussian. More explicitly, for some $\rho < 1/2$ and $C < \infty$, all sufficiently large n satisfy

$$\left|c^{-n} v_n(z)\right| \le e^{\rho|z|^2 + C|z|}.$$

The point-evaluation estimate $|f(w)| \le C_p e^{|w|^2/2}\|f\|_p$ now gives a uniform Gaussian majorant in the norm integral for the normalized operators in (5). Hence

$$\sup_n \|c^{-n} W^n_{v,az}\| < \infty.$$

Evaluation at zero gives the reverse obstruction $\|W^n_{v,az}\| \ge |c|^n$. This proves part 2 when a is nonreal.

The real boundary case

Let $s = a \in \mathbb{R} \setminus \{0\}$. A rotation of the variable preserves the Fock norm and reduces (4) to

$$v(z) = c\exp\left(Qz - \frac{1-s^2}{2}z^2\right), \qquad Q \in \mathbb{R}. \tag{7}$$

The nth normalized power is

$$J_n f(z) = \exp\left(Q_n z - \frac{1-s^{2n}}{2}z^2\right) f(s^n z), \qquad Q_n = Q\frac{1-s^n}{1-s}. \tag{8}$$

Lemma 1 (Gaussian strip norm)**.** *For the operators in* (8)*,*

$$\|J_n\|^p \asymp |s|^{-n}, \tag{9}$$

where both comparison constants are independent of n*.*

Proof. We use the elementary Fock–Carleson criterion: for a positive measure μ,

$$\int_{\mathbb{C}} |f(z)|^p e^{-p|z|^2/2}\, d\mu(z) \le C\|f\|_p^p \quad \Longleftrightarrow \quad \sup_{\zeta\in\mathbb{C}} \mu(D(\zeta,1)) < \infty, \tag{10}$$

with comparable constants. The necessity follows by testing normalized Fock kernels; the sufficiency follows from the submean inequality on a bounded-overlap disk covering.

Set $t = |s|^n$ and change variables $w = s^n z$ in the norm of $J_n f$. Relative to the Fock density $e^{-p|w|^2/2} dA(w)$, the new measure has density

$$\rho_n(w) = t^{-2}\exp\left(pQ_n \operatorname{Re}\frac{w}{s^n} - p\frac{1-t^2}{t^2}(\operatorname{Re} w)^2\right). \tag{11}$$

After completing the square, this is a Gaussian strip of width comparable to t and height comparable to t^{-2}; the completing-square factor stays bounded because (Q_n) is bounded and $t \le |s| < 1$. Therefore

$$\sup_\zeta \int_{D(\zeta,1)} \rho_n\, dA \asymp t^{-1}.$$

Equation (10) proves (9). $\square$

Since $W^n_{v,sz} = c^n J_n$, Lemma 1 yields

$$\|W^n_{v,sz}\| \asymp \big(|c|\,|s|^{-1/p}\big)^n. \tag{12}$$

This proves the sharp threshold in part 3 and completes the proof of Theorem 1.

5 Why the remaining zero-bearing boundary is genuine

The residual class is not empty. Fix real $0 < |s| < 1$, put $\beta = 1 - s^2$, and take $\tau \in \mathbb{R} \setminus \{0\}$. Then

$$v_\tau(z) = e^{-\beta z^2/2} \cosh(\tau z) \tag{13}$$

has infinitely many zeros. Moreover

$$|v_\tau(z)| e^{(|sz|^2 - |z|^2)/2} = |\cosh(\tau z)| e^{-\beta(\operatorname{Re} z)^2}$$

is bounded, so $W_{v_\tau, sz}$ is bounded. It is noncompact because the last display stays equal to one along a sequence on the imaginary axis. Multiplying v_τ by a sufficiently small nonzero scalar makes the operator a contraction, hence power bounded. Thus zeros cannot simply be excluded by a compactness or power-boundedness argument.

For a general zero-bearing boundary weight, the exact remaining condition can be written using

$$u_n(z) = \prod_{j=0}^{n-1} u(a^j z)$$

and uniform Fock–Carleson bounds for the pullback measures of $u_n(z) f(a^n z)$. Unlike the zero-free case, there is no quadratic logarithm that turns these products into a single Gaussian strip. Controlling this uniform family is the precise unresolved obstruction.

6 Eight focused upgrade attempts

1. **Later-literature closure.** Exact title, fixed-point, power boundedness, and Fock-space searches found the two 2021–2022 papers already cited by the source, but no later full classification.

2. **Weyl centering.** The isometric conjugacy (1) reduced the problem to $z \mapsto az$ without changing the decisive fixed-point value.

3. **Compact spectral route.** Taylor-leading terms and compact spectral theory removed the published zero-free hypothesis completely.

4. **Zero-free factorization.** Hadamard factorization reduced all zero-free bounded weights to quadratic exponentials and isolated the critical noncompact boundary.

5. **Nonreal boundary.** The strict inequality (6) made the normalized powers uniformly Gaussian and recovered the fixed-point criterion.

6. **Real boundary.** The Gaussian strip calculation produced the sharp $|a|^{1/p}$ threshold, closing the gap left by the earlier one-sided estimates.

7. **Boundary-zero elimination.** This route fails: the explicit family (13) is bounded, noncompact, and has infinitely many zeros; small scalar multiples are even contractions.

8. **Uniform product/Carleson route.** It gives an exact measure criterion for the full problem, but no scalar reduction for arbitrary zero sets. Nested products can concentrate on moving Gaussian strips, which is the remaining obstruction rather than a missing local bound.

7 Verification, novelty, and scope

The proof was checked at four sensitive interfaces: Weyl conjugacy preserves $v(0) = u(\alpha)$; the compact peripheral eigenvalue is semisimple; equality in $|1 - a^2| \geq 1 - |a|^2$ occurs exactly for real a; and the disk mass of (11) is uniformly comparable to $|s|^{-n}$. For $p = 2$, the same real-boundary exponent follows independently from the Bargmann transform: the canonical Gaussian dilation has norm $|s|^{-1/2}$.

A bounded search covered the run's four cheap indexes, the exact source and supporting papers, the parsed local corpus, and focused arXiv/web queries. No statement of Theorem 1 was found. The compact conclusion is close to consequences of known spectral descriptions, and Fock–Carleson methods are standard, so novelty confidence is moderate rather than definitive.

Human review should prioritize Lemma 1, especially the uniform Carleson constants at the equality threshold, and the zero-free boundedness classification used in (3)–(4). The packet does not claim a solution for noncompact weights having zeros.

where K is independent of n; hence uniform boundedness of the powers $W^n_{w,\varphi}$ on $\mathcal{F}^\nu$ is again equivalent to the condition $|w(0)| \le e^{-|b|^2/2}$. $\square$

The analysis is apparently intractable in the case $|a| < 1$, however we may consider the case when the weight w does not vanish, which will be of importance when we come to consider semigroups. In this case, writing $\beta = 1 - |a|^2$ we have the following result from [7].

Theorem 3.2. *For $\varphi(z) = az + b$ with $|a| < 1$ and w nonvanishing the operator $W_{w,\varphi}$ is bounded on $\mathcal{F}^\nu$ if and only if*

$$w(z) = e^{p+qz+rz^2}$$

and either:
(i) $|r| < \beta/2$, in which case $W_{w,\varphi}$ is compact on $\mathcal{F}^\nu$, or
(ii) $|r| = \beta/2$ and, with $t = q + \overline{b}a$, one has either $t = 0$ or else $t \ne 0$ and $r = -\frac{\beta}{2}\frac{t^2}{|t|^2}$. In case (ii) $W_{w,\varphi}$ is not compact on $\mathcal{F}^\nu$.

The analysis splits into two cases:

Case 1. We take $\varphi(z) = az + b$ with $0 < |a| < 1$, and $w(z) = e^{p+qz}$. That is, $r = 0$.

Here we once again have (11) with $w(0) = e^p$ but to check boundedness all we have is (10).

Case 2. We take $\varphi(z) = az + b$ with $0 < |a| < 1$, and $w(z) = e^{p+qz+rz^2}$ with $0 < |r| \le \beta/2$.

In Case 1, we now have

$$\begin{aligned} M_z(w_n, \varphi_n) &= |w_n(z)|^2 e^{|\varphi_n(z)|^2 - |z|^2} \\ &= |w(0)|^{2n} |e^{2q(z+az+\ldots+a^{n-1}z)}||e^{2qb(1+\frac{1-a^2}{1-a}+\ldots\frac{1-a^{n-1}}{1-a})}| e^{|\varphi_n(z)|^2-|z|^2} \\ &= e^{2n\operatorname{Re} p} e^{2\operatorname{Re}[qz(1-a^n)/(1-a)]} e^{2\operatorname{Re} qb[n/(1-a)-a(1-a^{n-1})/(1-a)^2]} \\ &\quad \times e^{|a^n z + \frac{1-a^n}{1-a}b|^2 - |z|^2} \end{aligned} \tag{12}$$

Taking logarithms, we have the following:

Theorem 3.3. *In Case 1:*
if $\operatorname{Re} p + \operatorname{Re}\left(\frac{qb}{1-a}\right) > 0$ then $W_{w,\varphi}$ is not power-bounded on $\mathcal{F}^\nu$;
if $\operatorname{Re} p + \operatorname{Re}\left(\frac{qb}{1-a}\right) - \ln|a| < 0$ then $W_{w,\varphi}$ is power-bounded on $\mathcal{F}^\nu$.

References

[1] I. Chalendar and J. R. Partington, *Semigroups of weighted composition operators on spaces of holomorphic functions*, arXiv:2201.09249 (2022), Section 6.2.2.

[2] I. Chalendar and J. R. Partington, *Weighted composition operators on the Fock space: iteration and semigroups*, arXiv:2106.00427 (2021), Section 3.

[3] W. Seyoum and T. Mengestie, *Spectrums and uniform mean ergodicity of weighted composition operators on Fock spaces*, Bull. Malays. Math. Sci. Soc. 45 (2022), 455–481; arXiv:2112.05369.